Product Description

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Client: Timothy Wang

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Product: 690 AM Radio Receiver

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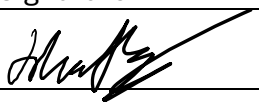
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1. Product Description History

1.1. Revision history

Changes Since Last Revision	Date	Version No
N/A	18/07/26	1.0
Removed PCB design information, as it has been discovered that its functionality was not properly verified. Removed case design information, as it is irrelevant without a proper PCB design. 1.1, 1.2, 1.3, 2.4, 3.1, 3.2, 3.3 have been updated to reflect the most recent changes accordingly.	06/10/26	1.1

1.2. Approval Signatures

Name	Signature	Title	Date	Version No
Timothy Wang		Project Manager	18/07/26	1.0
Timothy Wang		Project Manager	06/10/26	1.1

1.3. Distribution information

Name	Title	Date	Version No
Timothy Wang	Project Manager	18/07/26	1.0
Timothy Wang	Project Manager	06/10/26	1.1

2. Overview

2.1. Members

Name	Title
Timothy Wang	Project Manager

2.2. Stakeholders

Name	Relations/contribution to the project
Jasper Wang	Responsible for 3D printing the case
Dad	Responsible for funding the project, and giving design advice
CBC Radio One	Responsible for broadcasting 690 AM radio signals
Lee's Electronics	Responsible for supplying electrical components
SOLIDWORKS	Responsible for providing CAD software used for the case
KiCad	Responsible for providing CAD software used for schematic + PCB
Ltspice	Responsible for providing circuit simulation software

2.3. Purpose

Information about news and local traffic is often obtained through the TV and/or the internet. In the case of a power outage, these technologies may be unable to provide users with the most up to date news info. The 690 AM Radio Receiver can provide its users with the latest news during power outages, providing critical information when it would otherwise be unavailable. This product also has other use cases. For example, it may also be used during normal circumstances to get information about local weather and traffic. This project is also an exercise for me to further build on my skills and understanding in circuit building and applications of electromagnetism.

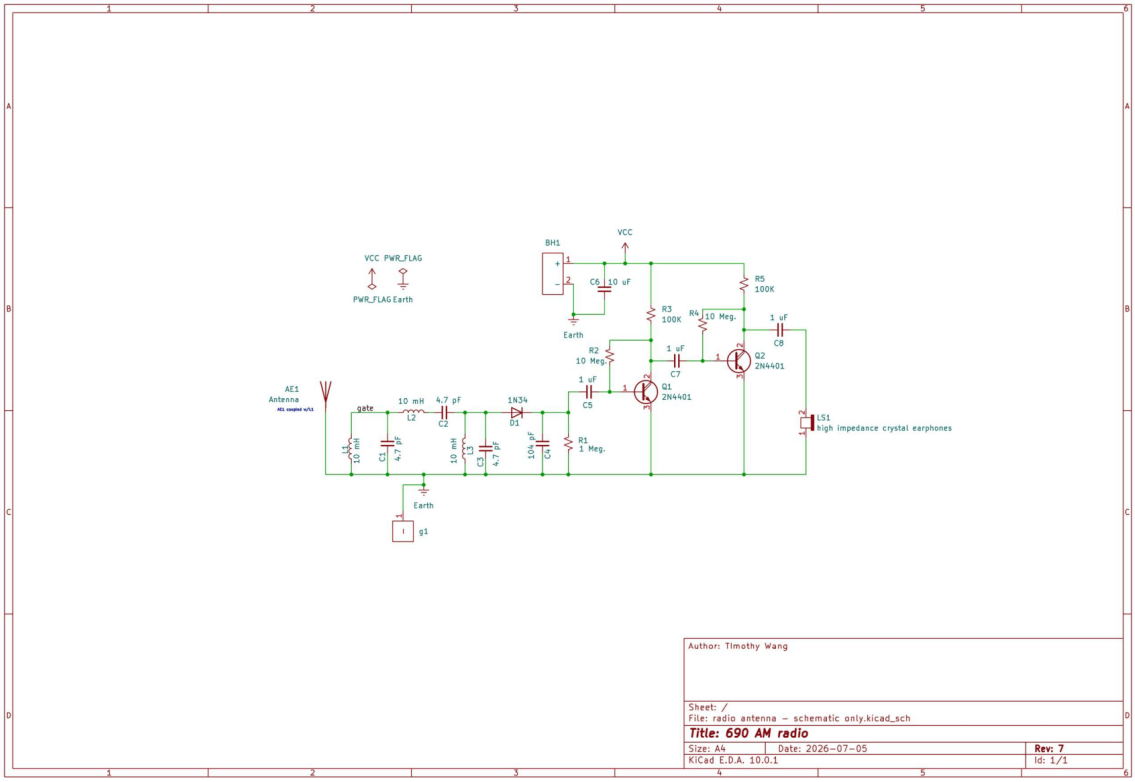
2.4. Product components

The 690 AM Radio Receiver product's circuit is comprised of these functional units:

1. Antenna
2. LC tuner circuit
3. AM demodulator
4. Amplifier stage
5. Power supplier
6. Earpiece

Listed below is the electrical schematic:

Electrical schematic:



Note: g1 is an extra ground reference, which was added so that an extra ground reference connection could be added to the PCB. Earth is used instead of GND to emphasize the importance of a good ground connection for the circuit. BH1 refers to a battery holder

3. Product Usage

3.1. Special considerations

This 690 AM Radio Receiver product is only able to pick up 690 AM broadcasts. Additionally, the battery which powers the radio requires manual attaching and detaching every time the user wants to turn on/off the radio. Depending on external environment conditions, more than one radio station may be heard.

3.2. Safety Information

- The battery is a hazard. Handle it with care.
- This product is intended to be powered by using a 9V battery.
- If the radio is too loud, discontinue use of the product.
- Do not operate this product near water. The inner electronics are fully exposed.

3.3. Instructions for use

1. Place the radio on a flat, stable surface.

2. Attach battery to battery holder.

3. Listen to the radio through the earpiece.